

RODRIGO LEITES-MENA

rodrigonleites1@outlook.com | (908) 372 - 8403 | Linden, NJ 07036

EDUCATION BS in Computer Science

9/2023-12/2027

Kean University, Union, NJ

- Anticipated Date of Graduation: 12/20/27
- Current GPA: 3.95

Technical Skills

Certifications

- Programming Languages: Java, Python, C, HTML, CSS, SQL
- Systems: Linux, xv6 kernel, Microsoft Azure Virtual Machines
- Tools: Visual Studio Code, Git, Eclipse, QEMU
- Google Cybersecurity Professional V2

Related Course Work

- Operating Systems, Data Structures and Algorithms, Computer Architecture, Linear Algebra, Discrete Math

PROJECTS

Caching Simulator

10/2025- 11/2025

- Built a C-based caching simulator that models hardware cache behavior using set associativity, valid bits, tags, and LRU replacement.
- Implemented the core useCache() function to handle cache hits, misses, and evictions using Least Recently Used (LRU) logic.
- Simulated cache operations on memory trace files and produced total hits, misses, and evictions across multiple configurations.
- Used GitHub Classroom for version control and development inside a Linux Azure Virtual Machine via remote SSH.
- Debugged and validated simulator correctness using the provided test-csim tool across 8 test cases.
- Applied concepts of caching, memory hierarchy, and systems-level C programming learned in Computer Architecture coursework.

Multi-Level Feedback Queue Scheduler in xv6

3/2025 – 4/2025

- Implemented a Multi-Level Feedback Queue scheduler in xv6, replacing the default round-robin scheduler.
- Completed the function find_index_of_max_wait() to schedule processes in the wait-ranking queue based on the largest waiting counter.
- Tested scheduler with multiple scenarios, verifying round-robin and wait-ranking behaviors, including demotion and promotion of processes.
- Built and ran xv6 on a Linux virtual machine using Visual Studio Code, GCC, Make, and QEMU
- Collaborated with baseline code provided via GitHub; used C for kernel modifications and scheduler implementation.
- Ensured system stability and prevented 100% CPU utilization bug by reviewing and fixing scheduling loops in proc.c

Azure Linux Virtual Machine Setup for xv6

2/2025–3/2025

- Provisioned a Linux VM on Microsoft Azure using free student credits.
- Cloned and compiled xv6 kernel, tested basic command-line operations and kernel programs.
- Managed VM resources efficiently, including stopping VM to save Azure credits.

WORK EXPERIENCE

BMW Product Specialist – Open Road BMW, Edison, NJ

8/2020–Present

- Diagnose and resolve complex technical issues, troubleshooting software and hardware efficiently.
- Acquire in-depth knowledge of BMW vehicles, including onboard computer systems and digital dashboards.
- Provide training and guidance on BMW operating systems and infotainment technology.
- Facilitate technical demonstrations using hands-on and virtual tools.
- Actively participate in ongoing technical education programs to stay current with BMW technologies.